



Research Paper

Numerical and experimental study on the influence of top bypass flow on the performance of plate fin heat exchanger



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HIGHLIGHTS

- Top bypass flow has great influence on thermal performance of PFHE.
- Experiments agree with simulations well with an average difference of 13.5%.
- A relationship between top bypass duct and its optimized air velocity is gained.
- A ratio of 0.5 for frontal fin to heat exchanger area is suggested.

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ABSTRACT

Bypass flow deserves a deep research as it is the important phenomenon in plate-fin heat exchanger and is of great influence on the thermal performance. Present studies had consulted this problem, but did not conduct a throughout study and restricted in a limited geometry. Therefore, in this paper an experimental bench is built for the research. Numerical simulations are also carried out on the models with and without bypass flow. The discrepancy of average heat transfer coefficient in simulation with bypass flow increases with air velocity and the maximum difference reaches 82.76% at the velocity of 10 m/s when compared to the model without that, but reduces to be 19.1% when compared to experimental results. After experimental validation, top bypass flow is analyzed with different ratios of top duct height to fin height and varied air velocities. A relationship between air velocity and its minimum height ratio is gained for the estimation in initial design of plate-fin heat exchanger. The optimization with varied fin numbers and interval distances is also carried out, and it is found that the optimized ratio of frontal fin area to frontal plate-fin heat exchanger area is almost keeping at an approximate value of 0.5 for all cases, but thinner fin and narrower air flowing channel can make better heat transfer effect. This paper reveals the influence of top bypass flow on thermal performance and design principle of plate-fin heat exchanger, which is hopeful to be an instruction for its future application in many fields.

1. Introduction

As a compact but high-efficiency component, plate-fin heat exchanger (PFHE) has been widely applied in chemical engineering, aerospace, electronics, power engineering and other fields, and is well beneficial to energy utilization, waste heat recovery, resource saving and cost reduction [1–3]. As a result, researchers have paid great attention to the developments on design theory, experimental test, manufacturing technique and newly application of PFHE, which will thereby make improvements of relevant fields mentioned above.

For PFHE, Kays et al. [4] did an extensive research on fifty-six types of the fins and gained their heat transfer and flow resistance curves and

relations. Gupta et al. [5] focused on heat loss problem in a crossing-flow PFHE due to low temperature and conduction in vertical direction. Sahin et al. [6] analyzed the effect of flowing structure on time-average and phase-average turbulence characteristic. Zhang et al. [7] built a distributed-parameter model to optimize the PFHE based on minimum entropy generation principle. Liu et al. [8] designed a new channel structure and using Fluent analysis they found that heat exchange efficiency of improved channels could be enlarged by enhancing internal turbulent flow when compared with traditional channels. Zhe et al. [9] experimentally investigated single-phase and two-phase flow distribution in PFHE and found that nonuniform distribution of fluid flow could affect system thermal performance. Cai et al. [10] demonstrated the

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performance of louver, straight, wavy and serrated fins at various altitudes by simulations and experiments, and found that louver fin was of the strongest cooling capacity while serrated fin was of the best adaptability to various altitudes.

According to these open-reported literatures, it can be observed that the design and performance analyses of PFHE have been treated analytically, numerically and experimentally and gained amount of achievements, but there are still many problems left and new challenges raised. Among of these challenges, bypass flow deserves a deep research as it is the important phenomenon in PFHE and is of great influence on the thermal performance.

The bypass flow phenomenon is always induced by a fan which distributes air through the parallel channels formed by plates fins and the baseplate as well as around the heat sink in the flow bypass region. That is that bypass flow is the possibility that the forced air can flow bypass the heat exchanger core entirely, and leak from the core to the bypass duct. Knight [11], Lee [12] and Teertstra [13] had studied particularly about the bypass flow, and accounted not only for flow by-pass (side bypass), but also for that part of the flow that enters the heat sink and exits through the top (top bypass). Furthermore, Chapman et al. [14] carried out a comparative thermal performance evaluation using aluminum heat sinks made with extruded fins, cross-cut rectangular pins, and elliptical shaped pins in environments characterized by a low air flow. The overall thermal resistance of a straight fin was lower than those of the other two designs owing mainly to the combined effect of enhanced lateral conduction along the fins and lower flow bypass characteristics where the heat source was localized at the center of the heat sink base plate. Butterbaugh et al. [15] investigated the effects of tip and lateral bypass on a heat sink with small fin spacing using compact modelling. Instead of balancing total energy (kinetic and pressure) of the fluid, they only balanced the pressure drop associated with the heat sink and bypass area in their iterative procedure to calculate channel velocity. The predictions of their flow model were within 10% of most of their experimental data. Hans et al. [16] developed an empirical bypass correlation for different fin designs. The Reynolds number and duct height had the largest influence on the prediction of Nusselt number while all investigated parameters (including Reynolds number, duct height, duct width, fin height, fin thickness and fin-to-fin distance) were important for dimensionless pressure drop. Dogruoz et al. [17] created a “two-branch by-pass model”, in which a one-dimensional difference approach was used to model the fluid flow through the heat sink and its top by-pass duct. It was shown that there was a good agreement between the temperature predictions based on the model and experimental data at high approach velocities for tall heat sinks, however the discrepancy increased as the approach velocity and heat sink height decreased. Rakib et al. [18] built an analytical model for predicting air flow and pressure drop across the heat sink by applying conservation of mass and momentum over the bypass regions and in the flow channels established between the fins of the heat sink. The simulated pressure drop varied significantly with bypass configurations (in terms of height, width and flowing velocity), and was found in good agreement with the experimental data. Villafane et al. [19] used numerical flow analyses to guide the design of the complete wind tunnel and analyze the reproduction of the bypass flow structure. Their study would lead to improvements in the thermal management of future aircraft power plants, particularly by taking benefit of the cold bypass air to refrigerate the lubrication oil, without penalizing the propulsive efficiency. Sangkeun et al. [20] conducted an efficient numerical procedure for the installation study of a cooler having a bypass duct to study the influence of an air–oil heat exchanger location and orientation on engine performance, and important design variables were clearly identified.

From these studies, bypass flow has been proved to affect thermal and flow characteristics of the PFHE remarkably, but there are still many problems needed solutions by numerical and experimental investigations. One of the problems is that how to design a suitable bypass

flow duct. For a suitable bypass flow duct, it firstly should be a duct for fully developed flow and consequently induce no additional contraction or expansion effect which can affect flow characteristics. Secondly, its dimensions should meet the requirements of compactness and lightweight. It should not be a big one which will enlarge system size and space greatly. And if for the determined bypass flow duct or system dimension, the coming problem is altered to be seeking the largest air velocity which is fit to these requirements. It can be concluded that there is a relationship between air velocity and its minimum bypass flow duct size. The present studies had consulted this problem, but they did not conduct a throughout study and only restricted in a limited geometry (a height range from 0.33 to 1 times of fin height for Hans [16], and a range from 1 to 2 times of heat sink height and width for Rakib [18]). Therefore, this paper will focus on the influence of bypass flow duct on thermal performance of the PFHE and a throughout study of the optimized ratio of bypass flow duct height to fin height for different air velocities. The numerical simulations and experimental tests are both carried out and their results are found in good agreement. After the validation, the PFHE is optimized with varied fin numbers and interval distances. This paper aims to reveal the influence of top bypass flow on PFHE performance and design principle of optimized PFHE, which is hopeful to be an instruction for future PFHE application in many fields.

2. Material and methods

Since the whole radiator is of a large size (referred to our work [21]), the PFHE presented here is just one part of the radiator due to calculation resource reduction and convenient establishment of experimental bench. In addition, its bottom surface is attached to another component closely, there is only top bypass duct left in flowing region so that the influence of top bypass flow is considered whereas side bypass flow is neglected in this work. The particular dimensions of PFHE are shown in Table 1. Subjected to limited lab conditions, an experimental model with a fixed fin number and thickness is applied in Section 2 and 3, whereas an actual model with different fin parameters is developed in Section 4 for a throughout study using the practical working conditions after experimental validation.

Table 1
PFHE analysis parameters and boundary conditions.

Material properties	Value	
	Experimental model	Actual model
Coolant fluid	Air	Air
Coolant viscosity (Pa s)	0.0000155	0.0000189
Coolant conductivity ($\text{W m}^{-1} \text{K}^{-1}$)	0.0263	0.0280
Coolant specific heat at 298 K ($\text{J kg}^{-1} \text{K}^{-1}$)	1005	1005
Coolant density (kg m^{-3})	1.768	1.110
Prandtl number Pr	0.702	0.699
PFHE material	Aluminum alloy	
PFHE conductivity ($\text{W m}^{-1} \text{K}^{-1}$)	180	
PFHE specific heat ($\text{J kg}^{-1} \text{K}^{-1}$)	871	
PFHE density (kg m^{-3})	2690	
Dimensions		
PFHE width W (mm)	100	100
PFHE length L (mm)	150	132
PFHE height H (mm)	45	49
Fin height h (mm)	35	35
Baseplate thickness $H-h$ (mm)	10	14
Fin thickness t (mm)	2	1/1.5/2/2.5
Fin number N	18	18–52
Boundary conditions		
Coolant inlet velocity (m s^{-1})	1–10	50
Coolant inlet temperature (K)	298	327
Coolant outlet pressure (Pa)	0	0
Uniform heat flux (W m^{-2})	3333	–
Uniform surface temperature (K)	–	413

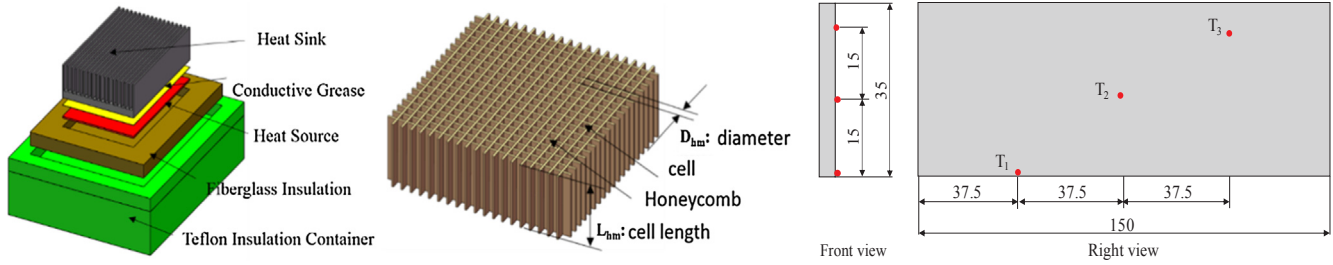


Fig. 1. Illustrations of tested PFHE (left), honey comb (middle) and thermometer distribution of fin (right).

2.1. Experimental apparatus and measurements

In order to test heat performance of PFHE, a small laboratorial wind tunnel was built. The wind tunnel walls were made of Plexiglas and its key frame was made of Aluminum alloy. The tested PFHE consisted of 18 aluminum fins with a thickness of 2 mm was placed in the middle of wind tunnel. A surface heater with a width of 100 mm, length of 150 mm and thickness of 3 mm and constant input power of 50 W, was placed at the bottom of baseplate to heat the PFHE. The silicone grease with a thermal conductivity of $4.5 \text{ W m}^{-1} \text{ K}^{-1}$ was smeared at the interface of heater and baseplate to lower contacted thermal resistance. Below heater plate, a fiberglass layer with a thickness of 5 mm and thermal conductivity of $0.035 \text{ W m}^{-1} \text{ K}^{-1}$ was used for insulation. The whole radiator was placed on a Teflon insulation platform with a thickness of 60 mm. Its illustration is shown in Fig. 1.

The air velocity was controlled by an axial fan with a maximum velocity of 20 m/s. The air from the axial fan firstly flew through an expansion section and a honey comb in the wind tunnel. The honey comb shown in Fig. 1 was set as a fairing component to straighten the flow into the test section and thereby gain a high-quality air flow. It was constructed with acrylic plate, 64% porosity ratio and 400 square holes. The cell length L_{hm} was of 56 mm and hole width D_{hm} was of 8 mm. Then the air passed through the tested PFHE and finally blew into atmosphere.

Night k-type sheathed thermocouples with an accuracy of 1.5 K were used to measure the temperatures in different positions of heat sink. They were distributed in third, ninth and fifteenth fin to measure fin surface temperatures, and each fin was with three thermocouples as shown in Fig. 1. The coolant air volume was measured by turbine flowmeters with an accuracy of 1%. The cooling air flow velocity was measured by industrial air velocity transmitter of Omega FMA1003A-MA with an accuracy of 1.5%. All the experimental data were collected by the testing modules of Yanhua Company. The schematic diagram and actual picture are shown in Figs. 2 and 3 respectively.

2.2. Numerical model establishment and analysis

The numerical simulation is carried out in Ansys Workbench and two models are developed. One is fully shrouded configuration in Fig. 4b, the other is without shroud and exposed in flowing air region

with top bypass duct in Fig. 4c. Its dimensions and working conditions are the same to experimental model shown in Table 1 of Section 2.

The FLUENT CFD solver is used to solve the governing equations for the conservation of mass, momentum and energy for a Newtonian incompressible fluid. This study only considers the steady state condition, and the relevant equations are summarized as:

$$\frac{\partial u_i}{\partial x_i} = 0 \quad (1)$$

$$\rho \frac{\partial (u_i u_j)}{\partial x_i} + \frac{\partial p}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\mu \frac{\partial u_j}{\partial x_i} \right) \quad (2)$$

$$\rho \frac{\partial (u_i c_p T)}{\partial x_i} = \frac{\partial}{\partial x_i} \left(\lambda \frac{\partial T}{\partial x_i} \right) \quad (3)$$

where u and x represent the velocity and direction vectors in Cartesian space; ρ is the density; p is the pressure; μ is the fluid viscosity; c_p is the specific heat capacity; T is the temperature; λ is thermal conductivity.

The simulation boundary is set by the following conditions:

1. Material of aluminum and air applied to the PFHE and coolant respectively.
2. Uniform inlet air velocity and its value of $1\text{--}10 \text{ m s}^{-1}$ but referenced pressure (atmosphere pressure) used at outlet plane.
3. Uniform and constant heat flux of 3333 W m^{-2} in bottom surface.
4. Constant inlet fluid temperature of 298 K and constant fluid properties at the temperature.
5. Adiabatic surface except heat transfer surface applied in the interfaces of air and PFHE.
6. 5 mm left for top bypass duct in simulation and experiment.

For continuity and momentum coupling, the solver uses Segregated/Implicit type, and adopts SIMPLE algorithm to solve the coupled analyses of speed and pressure. The type of discrete equation adopts second order upwind scheme. Convergence in all cases are declared only when both of strict criteria are satisfied: (a) reduction in all residuals of five orders of magnitude at least; (b) no observable change in surface temperature prediction for an additional tens of iterations.

Due to different air speeds in simulation, the flow will vary from laminar to turbulent status with air velocity increase. According to Reference 13, Re number is calculated by:

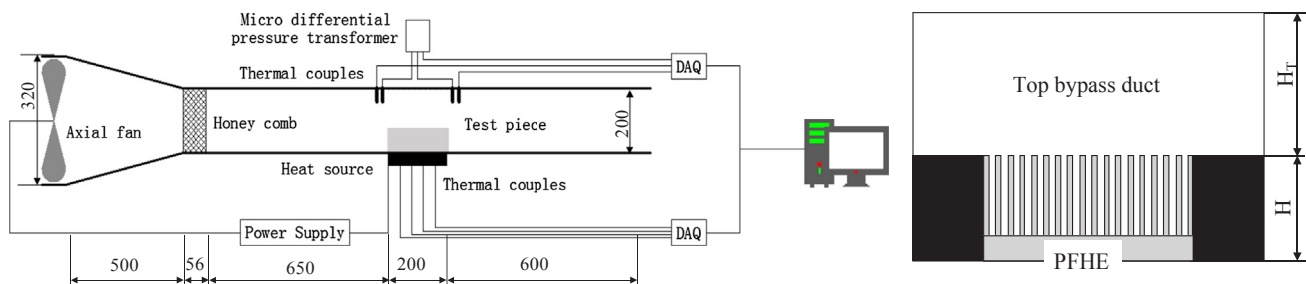


Fig. 2. Schematic diagram of experimental bench (left) and left view of PFHE (right).

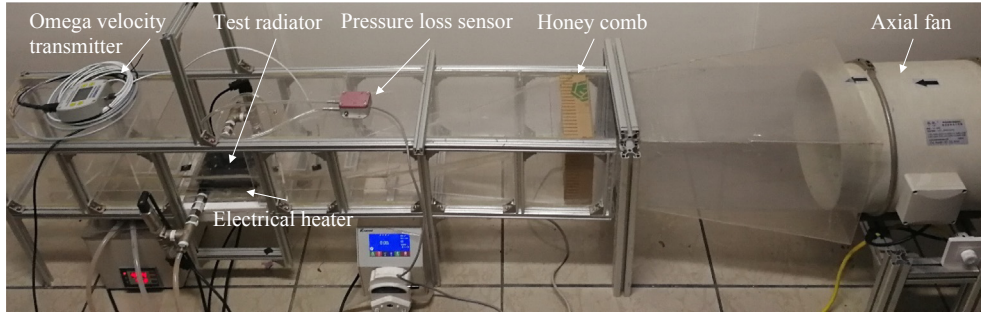


Fig. 3. Picture of experimental bench.

$$Re = b \frac{\rho v_{ch}}{\mu_a} \left(\frac{b}{L} \right) \quad (4)$$

where: b is fin interval distance, v_{ch} is air speed in fin channel and equals to $v_a(1 + t/b)$, v_a is incoming air speed, t is fin thickness, L is fin length (equal to PFHE length), μ_a is air viscosity. And from Ref. [22] the critical Re_c number is calculated by:

$$Re_c = 1.5946(\ln\alpha)^3 + 6.5771(\ln\alpha)^2 - 161.58(\ln\alpha) + 2195 \quad (5)$$

where: α equals to b/h , and h is fin height. If Re number is smaller than Re_c , it will set to be laminar flow in simulation, whereas if Re number is bigger than 4000, it will set to be turbulent and the turbulent flow will adopt k- ϵ model for the solution.

3. Numerical and experimental results

3.1. Thermal characteristics

The heat transfer mainly depends on heat conduction and convection from base plate to cooling air, which obeys heat transfer laws as follows:

$$q = -kA \left(\frac{dT}{dx} \right) \quad (6)$$

$$q = h_a A_{sa} \Delta T_{sa} \quad (7)$$

$$q = cm\Delta T \quad (8)$$

where: k is solid conductivity of PFHE, A is heat conduction surface, x is the direction along with fin height, c is solid heat capacity of PFHE, and ΔT is temperature variation of fin.

The average heat transfer coefficient of cooling air h_a is expressed as:

$$h_a = Q_{input} / (A_{sa} \Delta T_{sa}) \quad (9)$$

where: Q_{input} is heat power of bottom surface in PFHE and A_{sa} is fin surface area of the air. ΔT_{sa} is temperature difference which is defined by:

$$\Delta T_{sa} = T_{sw} - T_{in} \quad (10)$$

where: T_{sw} is average temperature of the fins, and T_{in} is surrounding temperature.

The fin surface area A_{sa} is defined by:

$$A_{sa} = N(2h \cdot t + 2h \cdot L) + W \cdot L \quad (11)$$

where: N is fin number, h is fin height, t is fin thickness, L is fin length (equal to PFHE length), W is fin width (equal to PFHE width).

Therefore, the thermal resistance of PFHE R_{sink} is defined by:

$$R_{sink} = \Delta T_{sa} / Q_{input} \quad (12)$$

3.2. Mesh independence study

The accuracy of a CFD analysis generally increases as the number of mesh elements increases. However, computing expense should also be considered, especially for optimization which requires hundreds of function evaluations (CFD analyses). Therefore, mesh generation should balance simulation reliability against computer calculation process and grid independence should be firstly performed to ensure the accuracy of simulation. For PFHE and flowing air domain, the structural grid is used as the model configuration is regular as shown in Fig. 5. Meshes of between 663,785 and 4,688,286 elements were generated. Two objective characteristics (average heat transfer coefficient and air pressure drop) were monitored during mesh convergence test. As shown in Fig. 6, when mesh number arrived to 3,252,104, the two values tended to be stable. The deviation value of average heat transfer coefficient for mesh number of 3,252,104 and 4,688,286 was 0.57%, and that of pressure drop was 1.29%. Therefore, in overall consideration of calculating efficiency and simulation error, total grid is set to be 3,252,104 elements in following simulations and the principles of mesh generation are executed in the other simulations.

3.3. Comparisons of numerical and experimental results

The comparisons of simulation and experimental results are shown in Fig. 7. The average heat transfer coefficient for simulation with fully shrouded is larger than that of no shroud and their discrepancy increases as the approach velocity increases. The maximum difference

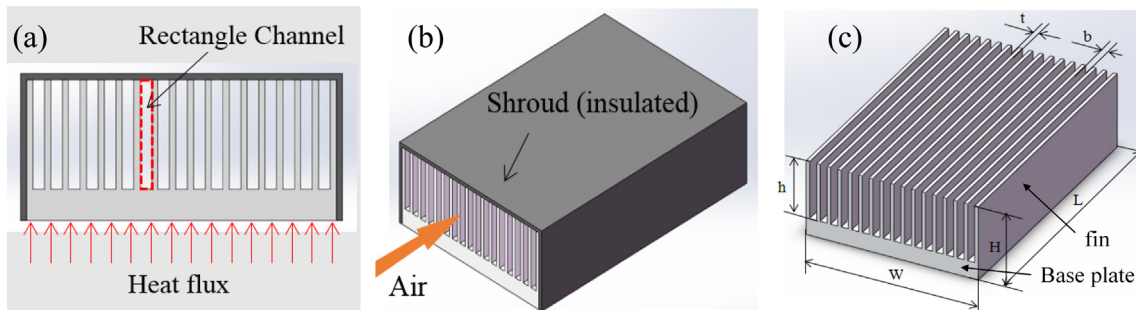


Fig. 4. Geometry illustration of simulation models.

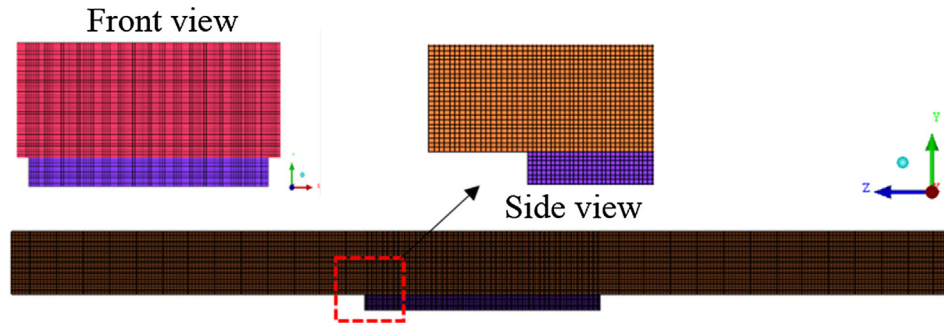


Fig. 5. Mesh generation of simulation domain.

reaches 82.76% when air velocity comes to be 10 m/s. That is because with fully shrouded, all cooling air is assumed to be forced into flowing channels so that its mass using in heat dissipation is larger than that of no shroud and hence its cooling capacity is stronger. With air velocity increase, the leak volume enlarges for the PFHE without shroud, leading to an increasing error between the two situations. Therefore, bypass flow can deteriorate heat dissipation performance, and should be considered in the design and analysis of PFHE, especially for a high approach velocity.

For the results of simulation and experiment without shroud, simulation result is a little larger than experimental result and their average difference of heat transfer coefficient is 13.5%. The main reason is that there is some heat loss from heat source, as the system cannot be fully adiabatic. Another reason is that there is still contact thermal resistance existing between electronic heating plate and radiator baseplate, although a high conductivity silicone grease is used to glue the two parts.

4. Top bypass flow analysis and PFHE optimization

According to the studies above, it is obvious that bypass flow plays a key role on thermal performance of a PFHE. Due to no side bypass flow here, only top bypass flow is analyzed in this section. After experimental validation of simulated reliability and accuracy, numerical research will be conducted on the actual PFHE model used the same simulated process and CFD method of experimental model. The actual model shown in Table 1 of Section 2 is with a few changes on the dimensions as it is averagely segmented from the large whole radiator. Instead of constant heat flux, thermal boundary condition is applied with constant temperature of 413 K as there is hot oil flowing in internal channel embedded in the baseplate.

4.1. Top bypass flow analysis

The actual model is shown in Fig. 8. The cooling air passes through flowing channels and top bypass duct and takes heat away from hot oil flowing in the internal channel of the baseplate. For an accurate prediction of actual heat transfer capacity and a high calculated efficiency, top bypass flow should be firstly analyzed to determine the height of top bypass duct with different air velocities. One fin model is established for thermal characteristic simulation with air velocity of 50 m/s and inlet temperature of 327 K, and the results are shown in Fig. 9. The height ratio is defined by the ratio of air domain height to fin height. As height ratio varies from 1 to 8, average heat transfer coefficient changes from 411.3 to 316.4 $\text{W m}^{-2} \text{K}^{-1}$, which is approximately 23% reduction and the great influence of top bypass flow is again proved. The average heat transfer coefficient reduces at first and then tends to be constant until height ratio reaches 8. With height ratio increase from 7 to 8, the value deviation is lower than 1%, which is the same to heat dissipation power also shown in Fig. 9. Therefore, for a PFHE with an air velocity of 50 m/s, top bypass duct should be at least 8 times higher than fin height, where the design analysis and numerical simulation can be regarded as accurate and reliable.

The minimum height ratios needed for different air velocities are also shown in Fig. 9. The relationship can be expressed by a curve-fitting equation:

$$r_{\min} = -0.00008c^3 + 0.0104c^2 - 0.2881c + 7 \quad (13)$$

where: r is the ratio of air bypass flow duct height to fin height, c is air approach velocity.

According to Eq. (13), an approximate height of top bypass flow duct and thereby the system size of the PFHE with given fin height and air velocity can be predicted. And if for a determined bypass flow duct or system dimension, the largest air velocity available for the PFHE can also be determined from the equations. This can be a very useful and

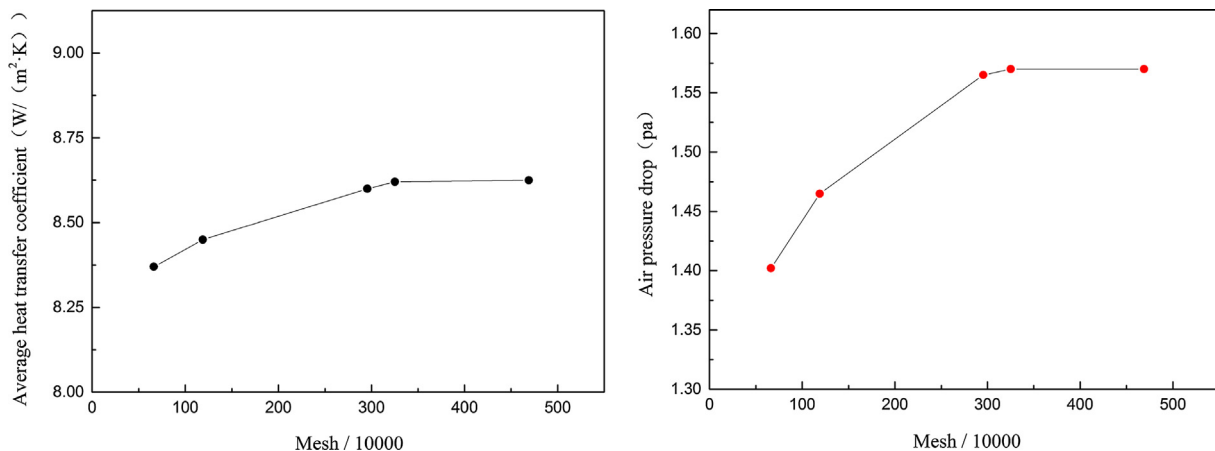


Fig. 6. Average heat transfer coefficient (left) and air pressure drop (right) vs. mesh numbers.

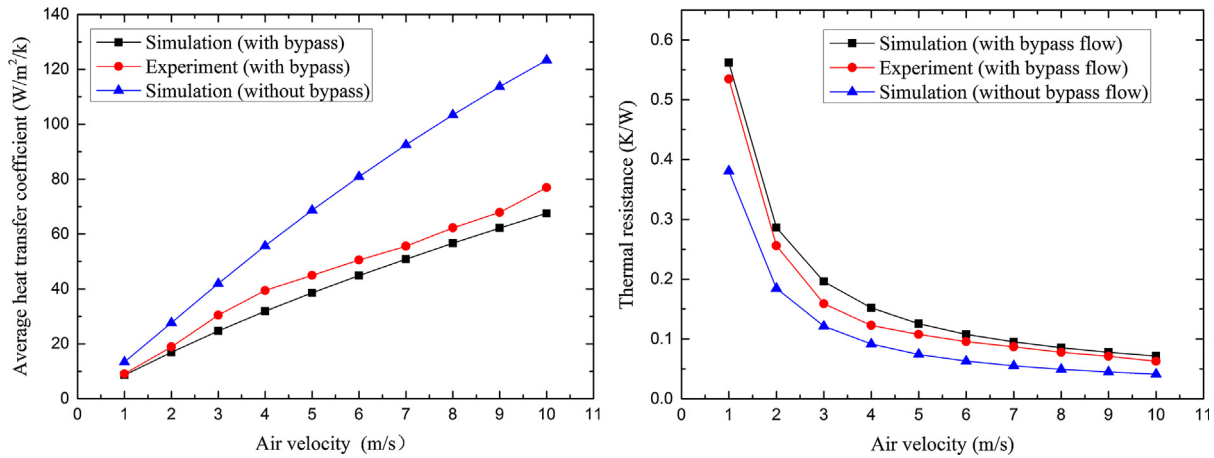


Fig. 7. Average heat transfer coefficient (left) and thermal resistance (right) vs. air velocity.

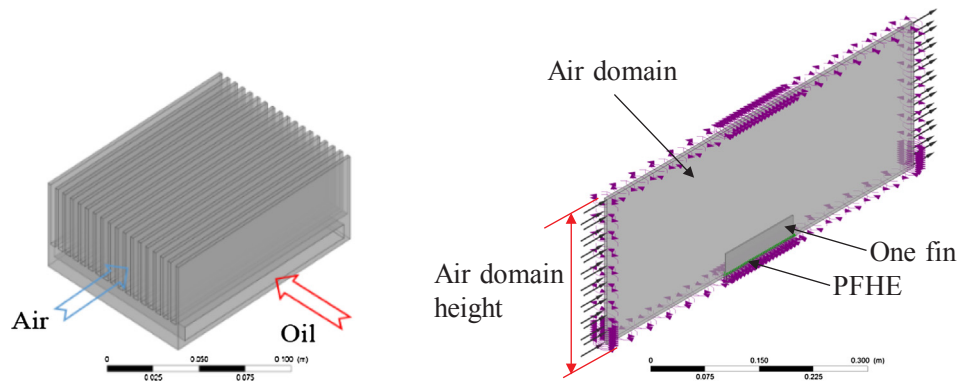


Fig. 8. Geometry illustration of actual model (left) and its simulation model with air domain (right).

instructive principle for the initial design of PFHE.

4.2. Optimization of the PFHE

After top bypass flow studies, the air domain height is fixed to be 8 times to fin height. In this setting, the optimization of PFHE is carried out by different fin numbers and fin thicknesses and their results are shown in Fig. 10. With fin thickness increase, the corresponding fin number of maximum average heat transfer coefficient decreases, but the value of maximum average heat transfer coefficient is almost the

same for the four fins. However, due to different fin surface areas A_{sa} , the maximum heat dissipation power and its corresponding fin number both raise with fin thickness decrease. The largest heat dissipation power of all cases comes to be 2.9 kW when the fin number is 44 and fin thickness is 1 mm. That is because in this case, its heat dissipation area is the largest of 0.432 m^2 . As a result, it can provide a maximum heat dissipation power among all cases. All results can also be consulted in Table 2.

Note that an interesting phenomenon is the ratio of frontal fin area A_{ff} (equals to $N \cdot t \cdot h$) to frontal PFHE area A_f (equals to $W \cdot h$) is almost

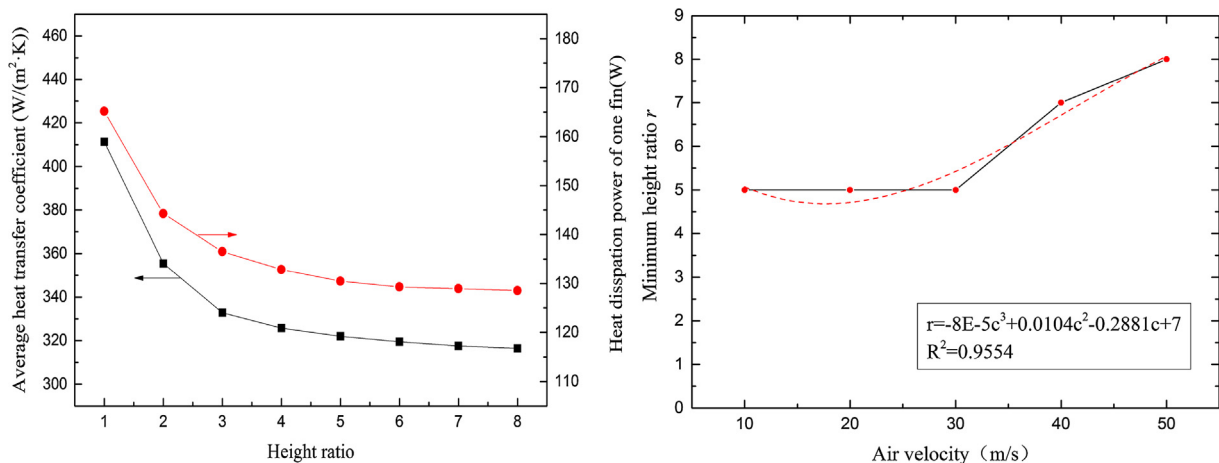


Fig. 9. Top bypass flow analysis: the left one is average heat transfer coefficient and heat dissipation power vs. height ratio for air velocity of 50 m/s; the right one is minimum height ratio vs. air velocity.

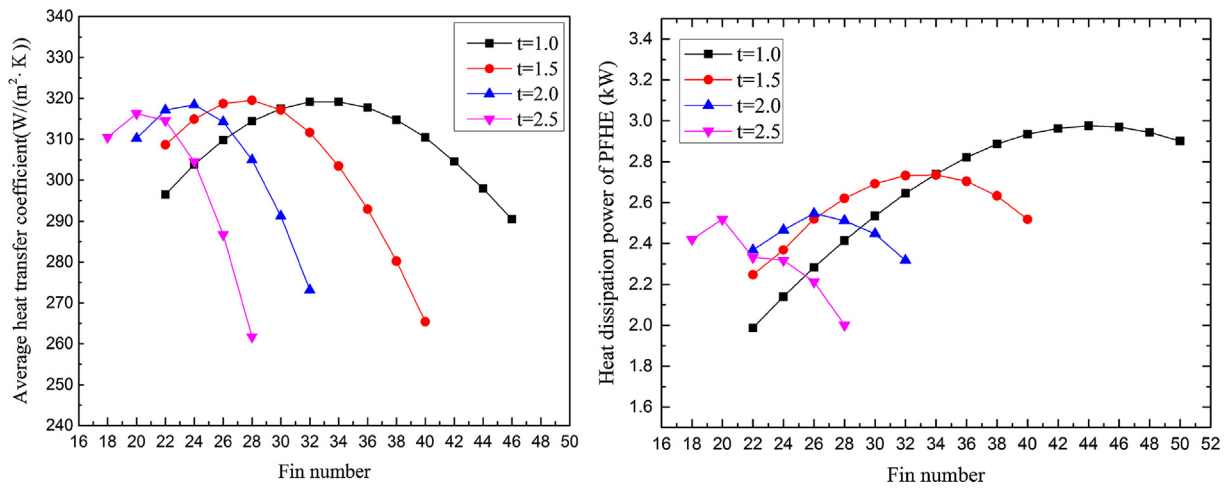


Fig. 10. Average heat transfer coefficient (left) and heat dissipation power (right) vs. fin number and thickness.

Table 2

Max thermal performance of different fin thickness and fin number.

Parameter	Value			
Fin thickness (mm)	1.0	1.5	2.0	2.5
Fin number of max average heat transfer coefficient	32	28	24	20
Max average heat transfer coefficient ($\text{W m}^{-2} \text{K}^{-1}$)	319.15	319.50	318.41	316.32
Fin number of max heat dissipation power	44	34	26	20
Fin surface area (m^2)	0.432	0.331	0.257	0.202
Max heat dissipation power (kW)	2.976	2.724	2.547	2.519
Frontal area ratio of fin to PFHE	0.44	0.51	0.52	0.5

keeping at an approximate value of 0.5. That is because for a fixed PFHE (with fixed frontal area A_f), the larger frontal fin area A_{ff} is, the smaller air flow area is. Consequently, there will be not enough air flowing into the channels to cool the fins and will lead to insufficient cooling capacity. On the converse, if frontal fin area A_{ff} is too small, there will be not enough dissipation area for heat conduction and convection and also will lead to bad heat dissipation in the heat sink. Hence the confirmation of fin parameters should balance the volume of cooling air and heat dissipation area. And in this work, to keep the frontal area ratio at an approximate value of 0.5 means a good thermal performance for a PFHE. This can be another useful and instructive principle for the initial design of PFHE.

5. Conclusions

This paper concentrates on the influence of top bypass flow on thermal performance of the PFHE. An experimental bench with a small laboratorial wind tunnel is built to test the heat performance. Numerical simulations are also carried out on the models with and without bypass flow. The discrepancy of average heat transfer coefficient in simulation with bypass flow increases with air velocity and the maximum difference reaches 82.76% at the velocity of 10 m/s when compared to the plate-fin heat exchanger without that, but reduces to be 19.1% when compared to experimental results. After accuracy and reliability validation of numerical simulation, top bypass flow analysis is conducted with different ratios of top duct height to fin height and different air approach velocities. A relationship between minimum height ratio and air velocity is gained for the estimation in initial design of plate-fin heat exchanger. The optimization with varied fin numbers and interval distances is also carried out, and it is found that the optimized ratio of frontal fin area to frontal plate-fin heat exchanger area is almost keeping at an approximate value of 0.5 for all cases, but

thinner fin and narrower air flowing channel can make better heat transfer effect. This paper reveals the influence of top bypass flow on thermal performance and design principle of plate-fin heat exchanger, which is hopeful to be an instruction for its future application in many fields.

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